

# PhysicsMate: Curriculum-Grounded Bengali Physics QA with Small Language Models

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## 1. Motivation

Bengali has around 230 million speakers, but it barely shows up in educational language-model research. For secondary-school science there is almost no question-answering resource tied to an actual curriculum, and the schools that would benefit most usually have weak internet and cheap hardware. We picked a concrete target: the NCTB Grade 9–10 Physics syllabus. The question we set out to answer was simple to state.

Can a small, curriculum-grounded dataset teach a small model enough Bengali physics to be useful offline on a normal laptop, and does that benefit change with model size?

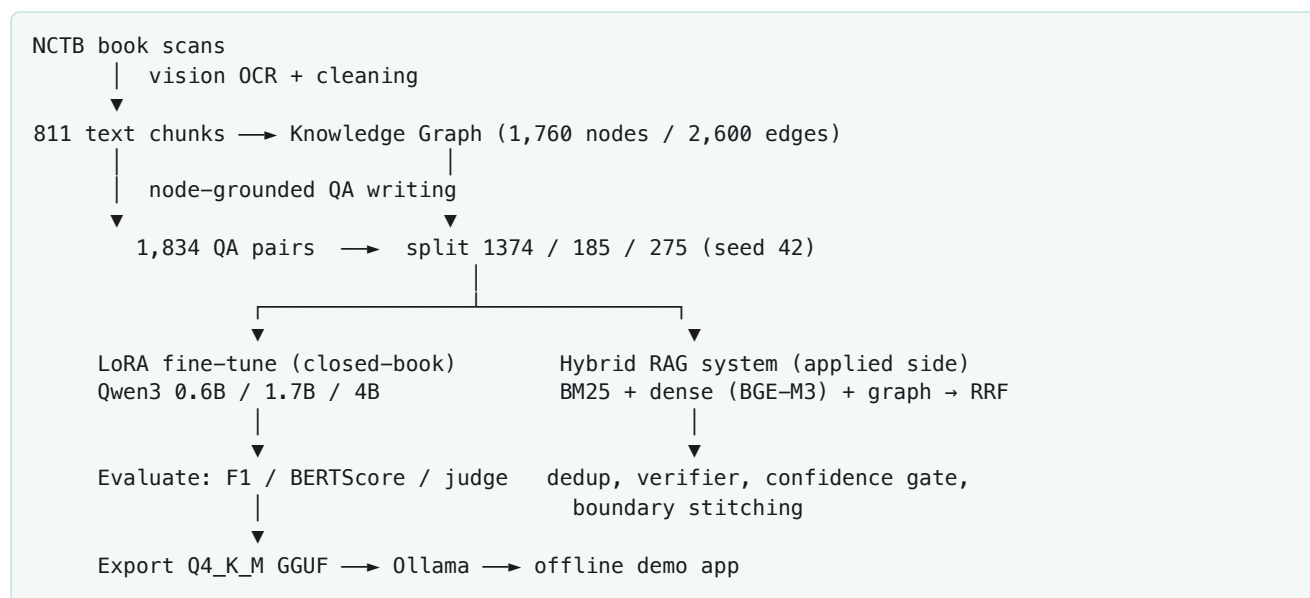
Two things made it worth doing. First, it gives us a benchmark for low-resource curriculum QA that other people can reuse. Second, it gives us something that actually runs: a fine-tuned model small enough to work with no internet, no API key, and no cloud bill. That last part matters, because that is the setting our users are in.

## 2. Methods (overview)

| Stage           | What we did                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data            | Digitised the NCTB Grade 9–10 physics book (page scans, vision OCR, per-chapter Markdown, then cleaning) into <b>811 text chunks</b> across <b>13 chapters</b> .             |
| Knowledge graph | Built a graph of <b>1,760 nodes</b> (10 node types) and <b>2,600 edges</b> to organise the domain.                                                                           |
| QA construction | Wrote <b>1,834 question–answer pairs</b> , each tied to a graph node, then split them (seed 42, stratified by chapter) into <b>train 1,374 / val 185 / test 275</b> .        |
| Adaptation      | LoRA / QLoRA fine-tuning of <b>4-bit Qwen3</b> at <b>0.6B, 1.7B, 4B</b> , closed-book (no retrieval), so the numbers reflect what the dataset itself puts into the weights.  |
| Evaluation      | Token-F1 and BERTScore, which are deterministic, plus an LLM-as-a-judge accuracy score, all on the held-out 275 questions.                                                   |
| Deployment      | Exported each model to a <b>Q4_K_M GGUF</b> file with an Ollama Modelfile, and built an offline demo (local server to Ollama, streaming, model switcher, Bangla/English UI). |

The main finding: fine-tuning helps at all three sizes, and the bigger the model, the bigger the jump (+5.5, then +15.0, then +23.3 accuracy points). Section 7 has the numbers.

### 3. System Overview



### 4. Weekly Progression

Reconstructed from file dates across the whole project ( ~/Downloads/transfer , ~/NCTB , ~/Desktop/conference paper ). It ran about ten weeks from the first OCR script to submission.

| Week | Dates (2026)    | Focus                                      | What got produced                                                                                                                                                                                                                                                         |
|------|-----------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W0   | Jun 19 – Jun 30 | First data pipeline (later revised)        | First OCR scripts; a clean-and-chunk pass; an early chapter-level graph; the first project concept note.                                                                                                                                                                  |
| W1   | Jul 1 – Jul 14  | First end-to-end prototype (later revised) | Dataset rebuilt a few times; QA parsed from the Udvash/SSC banks; a working retriever + agent + benchmark prototype.                                                                                                                                                      |
| W2   | Jul 21 – Jul 28 | Data foundation (final)                    | Book to Markdown, 13 chapters (Jul 21–23); figure extraction; per-chapter concept graphs; merged <b>dataset.json (811 chunks)</b> and <b>graph.json (1,760 nodes / 2,600 edges)</b> (Jul 25); <b>1,834 QA pairs</b> (Jul 26–28).                                          |
| W3   | Aug 1 – Aug 8   | Retrieval system                           | BM25 + dense index; hybrid retriever and fusion tuning; evaluation harness; the HSC (grade 11–12) bridge on the graph (Aug 8); first fine-tune attempts on Kaggle.                                                                                                        |
| W4   | Aug 9 – Aug 12  | Local fine-tuning + RAG work               | SFT dataset builder; local MLX LoRA fine-tune of Qwen3-4B; a 9-task RAG optimisation pass; a leakage-free "clean" dataset and an LLM judge.                                                                                                                               |
| W5   | Aug 13 – Aug 16 | Scaling study + paper                      | Closed-book split (1374/185/275); LoRA fine-tune of 0.6B/1.7B/4B; export to Q4_K_M GGUF + Modelfiles (Aug 13–14); deterministic evaluator, judge, and <code>verify.py</code> ; <code>results.yml</code> / <code>stats.yml</code> ; scaling figures; paper draft (Aug 16). |
| W6   | Aug 18 – Aug 24 | Deployment + write-up                      | Offline demo (three models through Ollama, streaming, Bangla/English) (Aug 18); UI fixes; this report and the slides (Aug 24–29).                                                                                                                                         |

## 5. Contribution of Each Team Member

| Member        | Main area                                | What they did                                                                                                                                                                                                                                                     |
|---------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rashid Azraf  | Dataset OCR & model tuning               | Ran the OCR that turned the Grade 9–10 book into clean text (the 811-chunk corpus); trained the LoRA/QLoRA models at 0.6B, 1.7B, and 4B; set up the closed-book training; exported the GGUF files and got them running in Ollama; ran the evaluation experiments. |
| Saadman Sajid | Synthetic QA, dataset analysis & writing | Wrote the 1,834 QA pairs and tied each to a graph node; did the dataset analysis (statistics, node-type and chapter breakdowns, the train/val/test design); read the related work; wrote the paper and all the reports.                                           |

Both of us worked on the knowledge graph, the demo app, final testing, and the presentation.

## 6. Implementation Details

*(Written up, not pasted. The source is in the repository.)*

### 6.1 Data construction

We turned the Grade 9–10 physics book into text one chapter at a time, reading each page image with a vision model and transcribing it into Markdown. We then cleaned out OCR junk and removed repeated text, which left 811 chunks.

On top of the text we built a knowledge graph. Every node has a type. The ten types and their counts are concept (397), question (252), quantity (224), figure (218), example (188), entity (176), definition (136), formula (83), law (65), and hsc\_topic (21). Nodes are joined by 2,600 typed edges, for example *defines*, *depends-on*, and *illustrated-by*. Each of the 1,834 QA pairs points back to a specific node, so we can trace any answer to a place in the curriculum. We split the data once, stratified by chapter with seed 42, into train, validation, and test (1,374 / 185 / 275). The test set never appears in training.

### 6.2 Fine-tuning

We adapt 4-bit Qwen3 at three sizes with LoRA. LoRA trains a small set of adapter weights and leaves the base model frozen, which is what lets the whole thing fit on one consumer GPU or an Apple-Silicon machine. We used two training paths over the project: a QLoRA path (rank 16, alpha 32, dropout 0.05, adapters on the attention and MLP projections, 3 epochs, cosine schedule, 8-bit AdamW, sequence length 512, seed 42) and an Apple MLX path for the local runs. Training is closed-book. The model only ever sees a question and its answer, with no retrieved context, so whatever it learns has to live in the weights.

### 6.3 Evaluation

We report three numbers on the 275 held-out questions. Token-F1 and BERTScore are deterministic, and `verify.py` recomputes them and checks they match the reported table. The third number is accuracy from an LLM judge, using the MT-Bench setup, where an answer counts as correct if it carries the gold answer's key fact. The judge scored all six model variants (raw and fine-tuned at each size) under the same prompt.

## 6.4 Deployment

Each fine-tuned model is exported to a Q4\_K\_M GGUF file (0.6B is 0.40 GB, 1.7B is 1.1 GB, 4B is 2.5 GB) with an Ollama Modelfile, so it runs offline through Ollama. A small local server, written with the Python standard library and no extra packages, serves a single web page and streams tokens from Ollama. The page lets you switch between the three sizes, streams the reply as it comes, offers sample questions from the real test set, handles greetings, and toggles between Bangla and English.

## 6.5 Applied RAG system

Next to the closed-book study we also built a retrieval system over the same corpus: BM25 for keyword match, BGE-M3 dense embeddings for meaning, and a graph channel, combined with reciprocal rank fusion. After retrieval the context goes through near-duplicate removal, derivation-boundary stitching, a second-pass groundedness check, and a confidence gate that refuses when the top scores are too low. On a separate leakage-free retrieval subset the fine-tuned 4B model came out even with RAG, which suggests the fine-tuning can take some load off retrieval at inference time.

## 7. Results

Closed-book, held-out test set (n = 275). Fine-tuned (FT) against the raw base:

| Model                | Params | Accuracy (judge) | Token-F1     | BERTScore    |
|----------------------|--------|------------------|--------------|--------------|
| Qwen3-0.6B raw       | 0.6B   | 0.0%             | 0.216        | 0.696        |
| Qwen3-0.6B <b>FT</b> | 0.6B   | <b>5.5%</b>      | <b>0.278</b> | <b>0.750</b> |
| Qwen3-1.7B raw       | 1.7B   | 10.5%            | 0.227        | 0.706        |
| Qwen3-1.7B <b>FT</b> | 1.7B   | <b>25.5%</b>     | <b>0.299</b> | <b>0.765</b> |
| Qwen3-4B raw         | 4B     | 27.6%            | 0.252        | 0.722        |
| Qwen3-4B <b>FT</b>   | 4B     | <b>50.9%</b>     | <b>0.350</b> | <b>0.782</b> |

The gain gets bigger as the model gets bigger:

| Model      | Accuracy raw → FT | Δ Accuracy      | Δ F1 | Δ BERTScore |
|------------|-------------------|-----------------|------|-------------|
| Qwen3-0.6B | 0.0% → 5.5%       | <b>+5.5 pp</b>  | +6.2 | +5.4        |
| Qwen3-1.7B | 10.5% → 25.5%     | <b>+15.0 pp</b> | +7.2 | +5.9        |
| Qwen3-4B   | 27.6% → 50.9%     | <b>+23.3 pp</b> | +9.8 | +6.0        |

On deployment: the fine-tuned 4B model, the best of the three, ships as a 2.5 GB file that runs offline on a laptop, and it matches RAG on the leakage-free subset.

## 8. Reproducibility & Deliverables

- Reproduce the deterministic numbers with `python verify.py`. It recomputes

Token-F1 and BERTScore for all six models and checks them against `results.yml`.

- To run the models or the demo, see `submission/README.md` . It has the terminal commands to load all three models in Ollama and 50-plus test questions.
  - Artefacts: the dataset and frozen split ( `data/` ), per-model predictions and GGUF files ( `models/` ), the evaluation scripts ( `evaluate.py` , `grade.py` , `verify.py` ), the paper draft ( `paper_section_*.md` ), and the demo ( `demo/` ).
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## 9. Limitations & Future Work

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The 0.6B model is still weak in absolute terms at 5.5%, so it is useful as a data point in the scaling story but not for a real classroom. The judge accuracy is an automatic proxy, and a human-rated subset would make it stronger. Next steps: push into HSC (grades 11–12) using the bridge already in the graph, add a human-verified test set, and try another model family such as Llama-3.2 to check the finding holds.